

Agenda

1. Announcements

- 11.4, 11.5, 11.6, 11.7 HW due today
- Discussion Quiz 8 tomorrow over ~~over~~
- Midterm 2 is next Tuesday April 11 at 7:30 pm

Exam Info Page: go.wisc.edu/1xxzfx

- Please fill out your TA evaluations

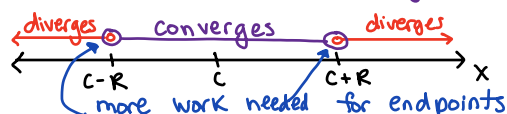
2. Section 11.8 - Power Series

3. Section 11.9 - Representing Functions as Power Series

Theorem: Suppose $\sum a_n(x-c)^n$. There are three possible cases:

- 1.) The series converges only at $x=c$. $R=0$ $\text{IOC: } \{c\}$
- 2.) The series converges everywhere $R=\infty$ $\text{IOC: } (-\infty, \infty)$
- 3.) For some $R>0$, the series converges for all x where

$|x-c| < R$ and diverges for all x where $|x-c| > R$.



$0 < R < \infty$ $\text{IOC: } (c-R, c+R)$
 possibilities $[c-R, c+R]$
 $[c-R, c+R)$
 $(c-R, c+R]$

The constant R is called the radius of convergence, and the interval for which the series converges is called the interval of convergence (IOC).

Example: Find the radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{(4x-12)^n}{(-2)^{2+n} n^2}$

$$\sum_{n=1}^{\infty} \frac{(4x-12)^n}{(-2)^{2+n} n^2}$$

Ratio Test!

$$\lim_{n \rightarrow \infty} \underbrace{\frac{(4x-12)^{n+1}}{(-2)^{3+n} (n+1)^2}}_{a_{n+1}} \cdot \underbrace{\frac{(-2)^{2+n} n^2}{(4x-12)^n}}_{\frac{1}{a_n}}$$

$$= \lim_{n \rightarrow \infty} \frac{\cancel{(-2)^{2+n}} (4x-12)^{n+1} n^2}{\cancel{(-2)^{3+n}} (4x-12)^n (n+1)^2}$$

$$= \lim_{n \rightarrow \infty} \left| \frac{(4x-12)^{n+1} n^2}{-2 (n+1)^2} \right| = \frac{|4x-12|}{2} \lim_{n \rightarrow \infty} \frac{n^2}{(n+1)^2}$$

$$= \frac{1}{2} |4x-12| \quad \begin{matrix} 4x-12=0 \\ x=3 \end{matrix}$$

$$= 2|x-3|$$

centred at $x=3$

Converges when $2|x-3| = 1$

$$\rightarrow |x-3| < \frac{1}{2}$$

$$\boxed{\frac{1}{2} = R}$$

$$\Rightarrow -\frac{1}{2} < x < \frac{1}{2} \rightarrow \frac{5}{2} < x < \frac{7}{2}$$

$$x = \frac{5}{2}: \sum_{n=1}^{\infty} \frac{(4(\frac{5}{2}) - 12)^n}{(-2)^{2+n} n^2} = \sum_{n=1}^{\infty} \frac{(-2)^n}{(-2)^{2+n} n^2} = \sum_{n=1}^{\infty} \frac{1}{4n^2} \quad \text{converges with } p=2 > 1$$

$$x = \frac{7}{2}: \sum_{n=1}^{\infty} \frac{(4(\frac{7}{2}) - 12)^n}{(-2)^{2+n} n^2} = \sum_{n=1}^{\infty} \frac{2^n}{(-1)^{2+n} 2^{2+n} n^2} = \sum_{n=1}^{\infty} \left| \frac{1}{(-1)^{2+n} \cdot 4 \cdot n^2} \right|$$
$$= \sum_{n=1}^{\infty} \frac{1}{4n^2} \quad \text{converges with } p=2 > 1$$

$$I_{OC}: \left[\frac{5}{2}, \frac{7}{2} \right]$$

Activity (If time): Determine the radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{3^n}{n} (x-4)^n$

Power Series as Functions

Recall geometric series: $\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots$

$a=1, r=x$ $\sum_{n=0}^{\infty} ar^{n-1}$
converges to $\frac{1}{1-x}$ on $(-1, 1)$
 $|r| < 1$

We can define a power series as

$$f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n \text{ on domain equaling IoC}$$

Examples: Write each function as a power series

$$1.) \frac{1}{1+x} = \frac{1}{1-(-x)} = \sum_{n=0}^{\infty} (-x)^n \quad -1 < x < 1$$

$$2.) \frac{x}{1-x} = \sum_{n=0}^{\infty} x \cdot x^n = \sum_{n=0}^{\infty} x^{n+1} \quad (-1, 1)$$

$$3.) \frac{1}{1-x^2} = \sum_{n=0}^{\infty} (x^2)^n = \sum_{n=0}^{\infty} x^{2n} \quad (-1, 1)$$

Activity: Write $\frac{2x^4}{1+x^2}$ as a power series

$$\frac{2x^4}{1-(-x^2)} \Rightarrow \sum_{n=0}^{\infty} 2x^4 (-x^2)^n = \sum_{n=0}^{\infty} -2x^4 x^{2n} \\ = \sum_{n=0}^{\infty} 2(-1)^n x^{2n+4}$$

Representing Functions as Power Series

Recall from Calculus I: $\frac{d}{dx}[x^n] = nx^{n-1}$ and $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

a_0 is c constant $\rightarrow a_0 + a_1(x-c) + \dots$

Theorem: Suppose we have the power series $f(x) = \sum_{n=0}^{\infty} a_n(x-c)^n$

with radius of convergence R . Then f is differentiable and integrable on the domain $(c-R, c+R)$ where:

1.) $f'(x) = \sum_{n=1}^{\infty} n \cdot a_n (x-c)^{n-1} = a_1 + 2a_2(x-c) + 3a_3(x-c)^2 + \dots$

2.) $\int f(x) dx = C + \sum_{n=0}^{\infty} a_n \frac{(x-c)^{n+1}}{n+1}$

Activity: $f(x) = \sum_{n=1}^{\infty} \frac{x^n}{n} = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots$ has IOC: $[-1, 1)$

1.) Find the IOC of $f'(x)$

2.) Find the IOC for $\int f(x) dx$